

Coordinate Geometry



Distance and midpoint formulas

- 1 Find the distance between $(2, -1)$ and $(10, 5)$.
 - 2 Find the midpoint of the segment connecting $(-4, 7)$ and $(6, -3)$.
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Slope and equations of lines

- 3 Find the slope of the line through $(1, 4)$ and $(5, -8)$.
 - 4 Write the equation of the line through $(2, -3)$ with slope $\frac{1}{2}$, in slope-intercept form.
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Parallel and perpendicular lines

- 5 Find the equation of the line parallel to $y = 4x - 1$ passing through $(0, 6)$.
 - 6 Find the equation of the line perpendicular to $y = -\frac{2}{3}x + 5$ passing through $(4, 1)$.
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The equation of a circle

- 7 Write the equation of a circle with center $(3, -2)$ and radius 7.
 - 8 Find the center and radius of the circle $x^2 + y^2 - 6x + 4y - 3 = 0$, by completing the square.
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Reflections, translations, and symmetry

- 9 Find the coordinates of the point $(5, -3)$ after a reflection over the x -axis, then after a reflection over the y -axis.
 - 10 A point (a, b) is translated 3 units right and 5 units down. Write the coordinates of the image point.
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Finding area using the coordinate plane

- 11 Find the area of the triangle with vertices $(0, 0)$, $(6, 0)$, and $(0, 4)$.

- 12 A rectangle has vertices $(1, 1)$, $(1, 5)$, $(7, 5)$, and $(7, 1)$. Find its area and perimeter.
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Determining if points are collinear

- 13 Determine whether the points $(1, 2)$, $(3, 6)$, and $(5, 10)$ are collinear, by comparing slopes.
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Finding the intersection of two lines

- 14 Find the point of intersection of $y = 2x - 3$ and $y = -x + 9$.
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Combining coordinate geometry concepts

- 15 This question has two parts. Triangle ABC has vertices $A(0, 0)$, $B(4, 0)$, and $C(2, 6)$. (a) Find the length of each side of the triangle. (b) Determine whether the triangle is isosceles, and justify your answer.
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Graphing inequalities in the coordinate plane

- 16 Determine whether the point $(3, 1)$ satisfies the inequality $y < 2x - 3$.
- 17 A region is defined by $y \geq x + 1$. Determine whether $(0, 0)$ lies in this region.
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The equation of a parabola in the coordinate plane

- 18 Write the equation of a parabola with vertex $(2, -3)$ that passes through $(4, 5)$.
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Finding points that divide a segment in a given ratio

- 19 Find the point that divides the segment from $(1, 2)$ to $(9, 10)$ in a ratio of $1 : 3$ (closer to the first point).
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A multi-part problem combining lines and circles

- 20 This question has two parts. A circle has equation $(x - 2)^2 + (y + 1)^2 = 25$. (a) State the center and radius. (b) Determine whether the point $(5, 3)$ lies inside, on, or outside the circle, by computing its distance from the center.

Systems of a line and a circle

21 Find the points of intersection of the line $y = x + 1$ and the circle $x^2 + y^2 = 25$.